Peter Yousefi

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SUMMARY

AI and Backend Software Engineer building production LLM applications and autonomous agent architectures in Python, TypeScript, FastAPI, and Next.js. Designs deterministic evaluation harnesses, Pydantic and Zod schema validators, and OpenTelemetry tracing pipelines backed by PostgreSQL and Azure cloud services to keep agent execution auditable, cost bounded, and under human approval. Combines enterprise SQL data reconciliation experience in live HR systems with automated testing and failure analysis.

EDUCATION AND CERTIFICATION

Bachelor of Computer Science, Trent University 2023 to 2026

Microsoft Certified: Azure Fundamentals | View credential

EXPERIENCE

Software Developer | Independent Projects Jun 2026 to Present

- Architected and deployed 7+ production AI agent and full stack web systems on Azure and Vercel using Python and TypeScript.
- Built CI/CD pipelines with Docker, GitHub Actions, and 560+ pytest, Vitest, and Playwright tests across all repositories.

Teaching Assistant, Trent University Sep 2025 to Dec 2025

- Graded 450+ Python and SQL assignments for 65 students with concept feedback, raising average code scores by 18%.
- Taught 65 students across 40+ hours of live Python and pytest debugging labs, cutting recurring bug rates by 35%.

HRIS Specialist (Intern), Concordia University Apr 2022 to Sep 2022

- Wrote 25+ multi table SQL queries and Python scripts across 4,500 employee records, cutting report time by 60%.
- Audited SQL data replication across 15,000+ ERP records during migration, fixing 120+ errors before payroll runs.

Vice President of Social, SCS Concordia Sep 2021 to Sep 2022

- Rebuilt the event schedule using SQL survey analysis and GitHub tracking across an 8 person team, lifting turnout 40%.
- Managed $15K+ budgets and vendor logistics in Excel for 10+ events of 200+ attendees, winning Event of the Year twice.

PROJECTS

Agent Flight Recorder | Execution control plane for AI agents Live demo

- Built an agent control plane in Python, FastAPI, and PostgreSQL to log 100% of run events, costs, and replay states.
- Added Azure Service Bus backoff retries, budget caps, and OpenTelemetry tracing across 195 pytest automated tests.

CrawlOps | Reliability checks for AI web research agents Live demo

- Built a TypeScript, Fastify, and Zod evaluator that grades OpenAI agent runs across 7 deterministic contract checks.
- Stored run telemetry in PostgreSQL and Prisma on Azure Container Apps, cutting regression review API costs by 100%.

AgentDrift | Detection of risky AI agent data movement Live demo

- Engineered a Python, FastAPI, and SQLAlchemy detector correlating reads and sends across 15 minute sliding windows.
- Tuned statistical rules on 380 holdout runs with pytest, lifting precision from 45.5% to 71.4% and cutting false alarms 67%.

AegisOps | Human approved AI incident response GitHub

- Built an Azure OpenAI, Python, and React workflow uniting 5 telemetry sources to cut incident triage time by 50%.
- Enforced Pydantic schemas and citation validation across 100% of LLM remediation plans in a PostgreSQL audit log.

Continuity | Consistent character storyboard generator Live demo

- Built a Next.js, TypeScript, and OpenAI API app splitting stories into 6 scenes with Zod validated character sheets.
- Moved image routes to Edge runtime with a 60s timeout and Playwright tests, cutting timeout failures by 90%.

Also: TerraScope | Falsifier (vets planet candidates via 7 Python tests) | Lumina (event driven campaign app using IBM Granite)

TECHNICAL SKILLS

AI / GenAI: LLM applications, OpenAI API, Azure OpenAI, structured output validation, evaluation harnesses, citation validation, agent observability

AI Software and Tools: Claude, Claude Code, OpenAI Codex, Cursor, GitHub Copilot, IBM Bob

Languages and Frameworks: Python, TypeScript, JavaScript, SQL, FastAPI, Fastify, Flask, Node.js, React, Next.js, Pydantic, Zod

Data, Cloud and Testing: PostgreSQL, Prisma, SQLAlchemy, Azure (Container Apps, Service Bus, Storage), Docker, GitHub Actions, OpenTelemetry, pytest, Vitest, Playwright